



Movable tray tilting setup

level, tray tilting status, heater status, humidifier status and exhaust fan status.

The logged data is graphically presented on a computer. In case of emergency, the requisite set of notifications is sent to the respective people.

Sakthi explains, "This system works by efficiently utilising a pre-charged battery's energy, rather than requiring a separate power generator. We have designed electronic measurement and control systems that assist in efficient use of the battery heater. This heater gives the same amount of power as a mains-powered heater would. A typical mains-powered heater operates off 230V AC supply, but a battery heater has to work from 12V DC supply. Hence the heater is designed using a suitable heating element that works off 12V DC supply. To avoid the use of an always-open vent (required to maintain air quality) as in commercial incubators, air quality is monitored to ensure that the vent is open only when required. This is again a big power saver."

Challenges

Karthik informs that the challenges were mainly three-fold: One related to the relative humidity measurement system, another concerning the battery heater and the third dealing with the mechanical system for the incubator.

Relative humidity. An IC-based sensor was initially used to measure the relative humidity. But it lacked robustness and reliability. The electronics embedded along with the sensing capacitive element was later found to be affected by the outside changing conditions.

To avoid the above problem, the electronics was separated from the capacitive element by a coaxial cable. The coaxial cable introduced a new problem of change in its

capacitance when the cable was bent for routing. The change in capacitance was very huge compared to small capacitive changes of the sensing element. As bending the cable for routing was unavoidable, the circuit had to be re-designed to suppress the capacitance of the coaxial cable. The circuit was designed and checked for reliability under extreme conditions. Humidity control is achieved by using an ultrasonic humidifier.

Battery heater. To maintain optimal temperature during power cut, the heater is designed to tap the stored electrical energy from the battery in the most efficient way possible. To convert the maximum-possible electrical energy stored in the battery into heat in the incubator, the heater is made to operate off 12V DC rather than 230V AC or 110V AC, which is the usual supply.

Karthik adds, "If we had to use a 230V AC heater even during power cut, it would have required conversion of 12V DC from the battery into 230V AC. Since the inverter efficiency is typically 50 per cent, the method of inverting to AC voltage for heater usage is an inefficient way. Hence a heater had to be designed that worked off 12V DC to avoid wastage of energy."

To get the same amount of energy from this battery heater as from a 230V AC heater, current flowing through the battery heater is scaled up to 19 times.

Mechanical system. Tray tilting is achieved using a linear actuator. The tilting system avoids vibrations and uses less power (20W). The tray tilting setup is made with wheels. Connections for power and feedback are provided through a single connector. The linear actuator is also mounted on the setup. Incubator walls are insulated using proper material to avoid any thermal leakage.

Prototype

The prototype was tested with around 5000 eggs. Though it cannot be looked upon as a permanent alternative to the diesel generator, it is definitely a much more economical solution for scheduled power cuts of around eight hours (10-12 hours with a 200Ah battery). During a scheduled power cut of two hours in morning, two hours in the afternoon and two hours in evening, it is, without doubt, a very apt solution as the battery also gets ample time to recharge itself.

"During our prototype test, we achieved almost 90 per cent hatch rate. We were also successful in lowering the electricity bill down by 50 per cent," said Sakthi.

Future additions

Right now, Karthik and Sakthi are working on getting this system patented. Talking about their future plans, they reveal, "We have got some excellent feedback. For the future, we have the idea of using all the data collected to get an insight into the hatching process. Maybe that's something we would like to do much later." ●

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